Open Source Software Audit Project

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Origin and Philosophy

The Problem: The "Token Ring" Crisis and the Student Rebellion

To understand why VLC exists, one must look back at the digital landscape of 1996, a time when the internet was still in its infancy and high-quality video was a luxury reserved for those with expensive, specialized hardware. As a student navigating the complexities of modern Computer Science, I find the audacity of the original VLC team—students just like us at Ecole Centrale Paris—to be a powerful reminder of how technical frustration can lead to global innovation.

The Technical Bottleneck of 1996

The primary "broken" element before VLC was the extreme cost and inaccessibility of video decoding. In the mid-90s, if you wanted to play a high-quality video file, your computer's CPU was generally too weak to handle the math required for decompression. You had to buy a physical MPEG decoder card, which could cost upwards of $1,000—a sum far beyond the reach of a typical college student.

Furthermore, the campus network at École Centrale Paris was built on a Token Ring architecture. While this was a standard networking technology at the time, it was not designed for the high-bandwidth, low-latency demands of streaming live television. The students wanted to watch satellite TV in their dorm rooms, but they faced two massive walls:

1. **Hardware Dependency:** There was no software-only player that could handle the stream.
2. **Network Incompatibility:** Existing tools were proprietary and could not be modified to work with the specific university network protocols.

The Birth of VideoLAN

Faced with these obstacles, the students did not wait for a corporate solution. They decided to build their own. This was the birth of the VideoLAN project. Their goal was radical: they wanted to decode video entirely through software, bypassing the need for expensive hardware.

Why Share? The 2001 Turning Point

The creators of VLC had a choice : they could have kept their code secret and tried to sell it to a tech giant like Microsoft or Apple , or they could share it with the world. They chose the latter for several philosophical and practical reasons :

- **Educational Growth :** The project started in a university environment where the sharing of knowledge was the core mission. Keeping it proprietary would have betrayed the collaborative spirit of the VIA Centrale Student Association .
- **The Power of Community :** By 2001, the project had grown beyond what a small group of students could maintain. On February 1, 2001, the director of École Centrale Paris authorized the release of VLC under the GNU General Public License (GPL) . This allowed developers from all over the world to contribute new codecs and features, turning it into the "Swiss Army Knife" of media players we use today .

In my view, the story of VLC is not just about a piece of software; it is about the "democratization of media". It proved that a group of students, driven by a desire to solve a local campus problem, could eventually break the hardware monopolies of the entire tech industry .

The License — What It Actually Says

To understand the legal DNA of VLC Media Player, I reviewed the license documentation provided by the VideoLAN organization . While VLC was originally released under the GNU General Public License (GPL) v2 , a significant portion of its core engine, libVLC, was re-licensed to the Lesser General Public License (LGPL) in 2011 to allow for wider integration .

The Four Freedoms of Free Software

The concept of "Free Software" is built upon four fundamental freedoms defined by the Free Software Foundation. After auditing the VLC license, I can confirm it grants all four:

- **Freedom 0:** The freedom to run the program for any purpose.
- **Freedom 1:** The freedom to study how the program works and change it so it does your computing as you wish. Access to the source code is a precondition for this.
- **Freedom 2:** The freedom to redistribute copies so you can help your neighbor.
- **Freedom 3:** The freedom to distribute copies of your modified versions to others. This gives the whole community a chance to benefit from your changes.

VLC’s GPL v2 license explicitly ensures these by requiring that anyone who receives the software also receives the right to these freedoms.

GPL vs. MIT Licensing: Which Would I Choose?

There is a profound philosophical difference between these two popular licenses:

Feature	GPL (Used by VLC/Linux)	MIT
Philosophy	**Reciprocal (Copyleft):** Focuses on the freedom of the *software*.	**Permissive:** Focuses on the freedom of the *user*.
Requirements	Modifications must be shared back if distributed.	No requirement to share modifications.
Commercial Use	Can be sold, but must remain open.	Can be sold as part of a closed, proprietary product.

My Choice: As a CSE student, if I built a tool meant to be a permanent community resource (like a security library), I would choose the GPL. It ensures that no one can take my hard work, improve it slightly, and then hide those improvements behind a paywall. However, if I wanted my code to be adopted as widely and easily as possible—even by big corporations—I would choose MIT.

'Free as in Beer' vs. 'Free as in Freedom'

This is perhaps the most misunderstood distinction in software.

- **Free as in Beer (Gratis):** Refers to the price. You can download VLC without paying a single Rupee.
- **Free as in Freedom (Libre):** Refers to liberty. It means you are not "locked in" by the developer.

The Ethics of Open Source: A Philosophical Reflection

Open source is far more than a legal framework or a way to distribute code; it is a profound statement on the democratization of knowledge. As a second-year Computer Science student, I see it as a digital "social contract" that balances individual contribution with collective progress. Below is a reflection on the ethical complexities that define the FOSS (Free and Open Source Software) movement.

Should All Software Be Open Source?

The debate over whether "all" software should be open is a conflict between the right to information and the right to property.

- **The Argument FOR:** Proponents argue that software has become the "infrastructure of modern life," much like roads or water. If the code governing our voting machines, medical devices, or banking systems is closed, we are forced to trust a corporation blindly. Open sourcing all software would eliminate "security through obscurity," allowing for a global audit of the tools that shape our reality.

- **The Argument AGAINST:** The counter-argument focuses on economic incentive and specialized innovation. If a small startup spends years of R&D and millions of dollars developing a breakthrough algorithm for cancer detection, open-sourcing it immediately might allow a larger competitor to copy it and drive the original innovators out of business. Without the ability to protect intellectual property, the financial motivation to solve "expensive" problems might disappear.

The Ethics of Profit on "Free" Labor

Companies like Google, Meta, and Red Hat have built trillion-dollar empires on top of Linux and other community-driven tools. Is this ethical?

- **The Parasitic View:** One could argue it is exploitative. Thousands of independent developers contribute to projects like VLC or the Linux Kernel for free, driven by passion or necessity. When a massive corporation takes that labor and wraps it in a paid service without directly compensating the original authors, it can feel like a violation of the "community" spirit.
- **The Symbiotic View:** However, these companies often provide the "industrial-strength" maintenance that volunteer communities cannot. Red Hat, for instance, pays hundreds of developers to work full-time on the Linux Kernel. By making the software viable for the enterprise, they create a massive ecosystem of jobs and stability that actually protects the long-term health of the open-source project.

Standing on the Shoulders of Giants

In software, "standing on the shoulders of giants" means that no developer starts from zero. Every line of code I write in my CSE labs sits on top of decades of open-source foundations—compilers, kernels, and libraries.

- **Enabling Innovation:** Open source acts as an accelerant. Because the "basics" (like the VLC engine or the Python language) are already built and shared, I can focus my energy on creating something new instead of reinventing the wheel.
- **Undermining Innovation:** Some argue it creates a "monoculture." If everyone uses the same open-source library, and that library has a flaw, the entire world is at risk. Furthermore, if a "good enough" open-source tool exists, there is less incentive for a company to build a "better" proprietary version, potentially slowing down radical breakthroughs.

Conclusion: A Personal Perspective

Reflecting on the "bane" of proprietary control—a topic I explored in my debate—I believe open source is the ultimate ethical safeguard. It ensures that while companies can profit, they cannot own the underlying logic of our world. As I continue my studies in C++, I see VLC not just as a player, but as a monument to the idea that knowledge is most powerful when it is shared.

Linux Footprint

1. Installation

- **Method:** Install via the terminal using the command: sudo apt install vlc.
- **Verification:** Confirm installation by running vlc --version.

```
patel_priyansh@LAPTOP-BOTQNC6B:~$  sudo apt install vlc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ vlc --version
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
```

2. Key Directories

Document these locations to show where the software "lives":

- **Binary (Executable):** /usr/bin/vlc.
- **User Configurations:** ~/.config/vlc/.
- **Logs:** System-level logs are found in /var/log/syslog.

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ /usr/bin/vlc
VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
[0000630962b7f220] main libvlc: Running vlc with the default interface. Use 'cvlc' to use vlc without interface.
[0000630962c15d90] main playlist: playlist is empty
```

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ ~/.config/vlc/
-bash: /home/patel_priyansh/.config/vlc/: Is a directory
```

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ /var/log/syslog
-bash: /var/log/syslog: Permission denied
```

3. Security & Ownership

- **User/Group:** VLC runs as the **current logged-in user** (e.g., your student username).
- **Reason:** This follows the "Principle of Least Privilege," ensuring the app cannot modify system files without permission.

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ ps -ef | grep vlc
patel_p+   26618     521  0 17:57 pts/0    00:00:00 grep --color=auto vlc
```

4. Service Management

VLC is managed as a process rather than a background system service:

- **Start:** vlc & (runs in background).
- **Monitor:** top or ps -ef | grep vlc.

- **Stop:** pkill vlc.

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ vlc &
[1] 26619
patel_priyansh@LAPTOP-BOTQNC6B:~$ VLC media player 3.0.20 Vetinari (revision 3.0.20-0-g6f0d0ab126b)
[00005ab7ea845220] main libvlc: Running vlc with the default interface. Use 'cvlc' to use vlc without interface.
[00005ab7ea8dbd90] main playlist: playlist is empty
^C
patel_priyansh@LAPTOP-BOTQNC6B:~$ ps -ef | grep vlc
patel_p+   26619     521  0 17:59 pts/0    00:00:00 vlc
patel_p+   26633     521  0 17:59 pts/0    00:00:00 grep --color=auto vlc
patel_priyansh@LAPTOP-BOTQNC6B:~$ pkill vlc
```

5. Update Model

- **Method:** Security patches are pushed via the community to official repositories.
- **Update Command:** sudo apt update && sudo apt upgrade.

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ sudo apt update && sudo apt upgrade
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

The FOSS Ecosystem

Technical Foundations (Dependencies)

VLC is famous for "playing everything," but it doesn't do it alone. It relies on a massive library of other open-source tools :

- **FFmpeg:** This is the "engine" under the hood. VLC uses FFmpeg's libraries (libavcodec) to decode almost every video and audio format you encounter .
- **Qt Framework:** The actual buttons, menus, and windows you see in the Linux version of VLC are built using the Qt toolkit .

Influence and Spin-offs

Because VLC's core engine (libVLC) is so powerful and open, it has inspired many other projects:

- **HandBrake:** A very popular open-source video transcoder that uses VLC's libraries to read and convert media files.
- **Mobile Apps:** Thousands of third-party media players on Android and iOS are simply "skins" built on top of the open-source libVLC engine .

Web and the "LAMP" Connection

While VLC isn't a traditional part of the LAMP stack (Linux, Apache, MySQL, PHP), it plays a huge role in the modern web :

- **Streaming Server:** VLC can act as a "mini-server" to stream live video over HTTP or RTSP, which can then be picked up by an Apache web server to be displayed on a website .
- **Web Plugins:** For years, the VLC web plugin was the primary way people watched high-quality video in browsers before HTML5 video became the standard .

Community and Governance

VLC is not owned by a company; it is governed by the VideoLAN Organization, a non-profit based in France .

- **Governance:** It is a volunteer-run project . Decisions are made by the core developers, many of whom have been with the project since its university days .
- **Communication:** The community stays connected through FOSDEM (a major European FOSS conference), the official VideoLAN forums, and Freenode IRC channels where developers discuss code in real-time .

Open Source vs Proprietary

To complete my audit of VLC Media Player, I compared it against its most prominent proprietary competitor, CyberLink PowerDVD. While both serve the primary purpose of playing high-quality media, their underlying philosophies create a very different user experience.

Comparison Table

Dimension	Open Source (VLC Media Player)	Proprietary Alternative (PowerDVD)
Cost	Completely free ($0).	Paid software; requires a one-time purchase or subscription (approx. $50–$100).
Security (Auditability)	**Transparent:** Any developer or security researcher can audit the source code for backdoors or vulnerabilities.	**Closed:** Only the company's internal developers can see the code; users must trust the corporate security team.
Support and Reliability	**Community-driven:** Reliable via forums and FAQs, but no "official" help desk.	**Corporate:** Provides professional 24/7 customer support and guaranteed stability for paying customers.
Freedom to Modify	**High:** Under the GPL/LGPL, users can modify the code to add a specific codec or change the interface.	**None:** Users are legally forbidden from modifying or reverse-engineering the binary files.
Control	**Community Control:** Governed by the non-profit VideoLAN Organization.	**Corporate Control:** Controlled by a for-profit entity focused on shareholder value and DRM compliance.

Critical Analysis: My Findings

After digging into both options, I realized that "Open Source" isn't a magic word for "better." For example, PowerDVD often wins in the niche area of high-end, licensed hardware support. Because PowerDVD is a commercial product, they can afford the expensive licensing fees required to play protected 4K Ultra HD Blu-ray discs natively—something VLC

struggles with because the community refuses to include proprietary decryption keys that violate their open-source principles.

However, for almost every other scenario, VLC is the clear winner for me as a CSE student. While PowerDVD is packed with heavy "upselling" features and telemetry that tracks your viewing habits, VLC is a lightweight, honest tool. It doesn't "phone home" or try to sell me a subscription. It just does the one thing it was built for : playing the file I gave it.

Overall Verdict

Given everything I have learned during this audit , I would absolutely recommend VLC Media Player for a real-world deployment , especially in educational or privacy-conscious environments . Its ability to run on virtually any hardware without licensing costs makes it an unbeatable foundation for a large-scale rollout. In a professional setting, the lack of a "corporate support line" is easily offset by the massive community knowledge base that already exists online.

As for contributing, this project has actually changed my perspective. Before this audit, I saw VLC as a finished "product." Now, I see it as a living, breathing codebase that needs fresh eyes. Since I am currently studying C++ and have an interest in network security, I would love to contribute by auditing one of the smaller, less-used decoder modules or helping with documentation for the libVLC engine. It feels like the right way to give back to a tool that has supported my own digital life for years.

Shell Script Tasks

Script 1 — System Identity Report

Code:

```bash
#!/bin/bash
# Script 1: System Identity Report
# Author: [Your Name]
# Course: Open Source Software

# --- Variables --- [cite: 105]
STUDENT_NAME="[Patel Priyansh]"
SOFTWARE_CHOICE="VLC Media Player"
KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DISTRO=$(hostnamectl | grep "Operating System" | cut -d: -f2)
DATE_TIME=$(date)

# --- Display --- [cite: 113]
echo "=========================================="
echo "        Open Source Audit: $SOFTWARE_CHOICE"
echo "        Student: $STUDENT_NAME"
echo "=========================================="
echo "Kernel Version : $KERNEL"
echo "Current User   : $USER_NAME"
echo "Home Directory : $HOME"
echo "System Uptime  : $UPTIME"
echo "OS Distribution:$DISTRO"
echo "Current Date   : $DATE_TIME"
echo "------------------------------------------"
echo "Notice: This OS is covered by Open Source licenses (e.g., GPL/MIT)."
echo "=========================================="
```

Output:

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ nano script1.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ chmod +x script1.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ ./script1.sh
==========================================
        Open Source Audit: VLC Media Player
        Student: [Patel Priyansh]
==========================================
Kernel Version : 6.6.87.2-microsoft-standard-WSL2
Current User   : patel_priyansh
Home Directory : /home/patel_priyansh
System Uptime  : up 5 minutes
OS Distribution: Ubuntu 24.04.1 LTS
Current Date   : Sun Mar 29 17:26:09 UTC 2026
------------------------------------------
Notice: This OS is covered by Open Source licenses (e.g., GPL/MIT).
==========================================
```

Explaination:

The goal of this script is to act as a "welcome screen" for a Linux system. It automatically gathers and displays essential metadata about the environment you are using for your audit.

Script 2 — FOSS Package Inspector

Code:

```bash
#!/bin/bash
# Script 2: FOSS Package Inspector
PACKAGE="vlc"

# Check if package is installed (using dpkg for Debian/Ubuntu or rpm for RHEL) [cite: 134, 135]
if command -v $PACKAGE &>/dev/null; then
    echo "SUCCESS: $PACKAGE is installed."
    # Display version and license info [cite: 139]
    apt-cache show $PACKAGE | grep -E 'Version|License|Description' || rpm -qi $PACKAGE | grep -E 'Version|License|Summary'
else
    echo "ERROR: $PACKAGE is NOT installed."
fi

# Case statement for philosophy notes [cite: 141]
case $PACKAGE in
    vlc)
        echo "Philosophy: VLC represents the freedom to play any format, born from student innovation." ;;
    httpd)
        echo "Philosophy: Apache: the web server that built the open internet." ;;
    mysql)
        echo "Philosophy: MySQL: open source at the heart of millions of apps." ;;
    *)
        echo "Philosophy: Supporting the FOSS ecosystem through transparency." ;;
esac
```

Output:

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ nano script2.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ chmod +x script2.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ ./script2.sh
SUCCESS: vlc is installed.
Version: 3.0.20-3build6
Description-en: multimedia player and streamer
Description-md5: d96da6b991aa50928533b16fb11b3136
Philosophy: VLC represents the freedom to play any format, born from student innovation.
```

Explaination:

The goal of this script is to verify if your chosen software (VLC) is currently installed on the Linux system. It serves as an automated audit tool that identifies the software version and explains its importance in the open-source world.

Script 3 — Disk and Permission Auditor

Code:

```
#!/bin/bash
# Script 3: Disk and Permission Auditor
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------------------------"

for DIR in "${DIRS[@]}"; do
    if [ -d "$DIR" ]; then
        # Extract permissions, owner, and size [cite: 157]
        PERMS=$(ls -ld "$DIR" | awk '{print $1, $3, $4}')
        SIZE=$(du -sh "$DIR" 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERMS | Size: $SIZE"
    else
        echo "$DIR does not exist on this system." [cite: 161]
    fi
done

# TODO: VLC specific config check [cite: 162]
VLC_CONFIG="$HOME/.config/vlc"
if [ -d "$VLC_CONFIG" ]; then
    echo "VLC Config Directory found at $VLC_CONFIG"
    ls -ld "$VLC_CONFIG"
fi
```

Output:

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ nano script3.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ chmod +x script3.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ ./script3.sh
Directory Audit Report
----------------------------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.4M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 78M
/home => Permissions: drwxr-xr-x root root | Size: 120K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 98M
/tmp => Permissions: drwxrwxrwt root root | Size: 24K
VLC Config Directory found at /home/patel_priyansh/.config/vlc
drwxr-xr-x 2 patel_priyansh patel_priyansh 4096 Mar 29 17:59 /home/patel_priyansh/.config/vlc
```

Explaination:

The goal of this script is to perform a security and storage audit of the Linux filesystem. It automatically checks a list of critical system directories to report their storage usage and their access permissions, ensuring that sensitive areas like /etc or /var/log are properly configured.

Script 4 — Log File Analyzer

Code:

```
#!/bin/bash
# Script 4: Log File Analyzer
# Usage: ./script4.sh /var/log/syslog "vlc"

LOGFILE=$1
KEYWORD=${2:-"error"} # Default keyword is 'error' [cite: 171]
COUNT=0

if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found." [cite: 176]
    exit 1
fi

# Read file line by line [cite: 177, 178]
while IFS= read -r LINE; do
    if echo "$LINE" | grep -iq "$KEYWORD"; then
        COUNT=$((COUNT + 1))
        # Store last match for tail-style output [cite: 184]
        LAST_MATCHES+="$LINE\n"
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"
echo "Last 5 matches:"
echo -e "$LAST_MATCHES" | tail -n 5
```

Output:

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ nano script4.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ chmod +x script4.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ ./script4.sh /var/log/syslog vlc
Keyword 'vlc' found 0 times in /var/log/syslog
Last 5 matches:
```

Explaination:

The goal of this script is to automate the process of searching through system log files to identify specific events. By searching for keywords like "ERROR" or "vlc," you can quickly audit how your software is behaving on the Linux system without manually reading thousands of lines.

Script 5 — The Open Source Manifesto Generator

Code:

```
#!/bin/bash
# Script 5: Open Source Manifesto Generator

echo "Answer three questions to generate your manifesto."
echo

read -p "1. Name one open-source tool you use every day: " TOOL
read -p "2. In one word, what does 'freedom' mean to you? " FREEDOM
read -p "3. Name one thing you would build and share freely: " BUILD

DATE=$(date +'%d %B %Y')
OUTPUT="manifesto_$(whoami).txt"

# Compose and write to file [cite: 199, 200]
echo "--- My Open Source Manifesto ---" > "$OUTPUT"
echo "Created on: $DATE" >> "$OUTPUT"
echo "By using $TOOL, I embrace the power of community." >> "$OUTPUT"
echo "To me, software freedom means $FREEDOM." >> "$OUTPUT"
echo "In this spirit, I commit to building $BUILD for the world to share." >> "$OUTPUT"

echo "Manifesto saved to $OUTPUT"
cat "$OUTPUT"
```

Output:

```
patel_priyansh@LAPTOP-BOTQNC6B:~$ nano script5.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ chmod +x script5.sh
patel_priyansh@LAPTOP-BOTQNC6B:~$ ./script5.sh
Answer three questions to generate your manifesto.

1. Name one open-source tool you use every day: VLC Media Player
2. In one word, what does 'freedom' mean to you? Innovation
3. Name one thing you would build and share freely: Useful Software Tools
Manifesto saved to manifesto_patel_priyansh.txt
--- My Open Source Manifesto ---
Created on: 29 March 2026
By using VLC Media Player, I embrace the power of community.
To me, software freedom means Innovation.
In this spirit, I commit to building Useful Software Tools for the world to share.
```

Explaination:

This is the most creative script in your project. Its goal is to allow a user to interactively generate a personalized open-source philosophy statement. By answering three simple questions, the script composes a formal "manifesto" that reflects the user's values and saves it as a permanent text file.

Conclusion

After completing this structured audit, my final verdict is that VLC Media Player is an essential piece of software for any real-world Linux deployment. Its "Linux footprint" is incredibly clean, and its ability to function without the licensing fees or "locked" source code found in proprietary tools like PowerDVD makes it the superior choice for both students and professionals. The fact that it grew from a small university project into a global standard proves that community-driven development can actually outlast and outperform corporate products.

This project has changed how I view the software I use every day. Open source isn't just about "free" software—it’s about the freedom to study, change, and improve the tools we rely on. Moving forward in my computer science career, I would definitely contribute to projects like VLC, as they represent the true spirit of "standing on the shoulders of giants". Understanding the philosophy behind these tools has given me a much stronger foundation for being a professional developer.